

Module 5: Bost Sum Formula Audit Report (v2)

Battle Plan v1.6 | David Fox | May 21, 2026

Supervisor correction applied: $S4 = \{2, 3, 19, 191\}$

STATUS: HALTED -- Formula discrepancy persists after supervisor correction.

$\text{arb_gt}(C, \text{threshold}) = 0$. Input chain intact. Awaiting formula correction.

1. Supervisor Correction Applied

The supervisor has clarified (v2) that Module 5 uses $S4$, not $S14$:

$S4 = \{2, 3, 19, 191\}$ (first 4 elements of $S(\alpha_0)$)
 $S4$ is a subset of $S14$, which was certified in Module 4.
Module 3 values $Q_5=226$ and 82829 are NOT used here.

The LaTeX claim is:

$C(S4) := \sum_{p \in S4} \log(p) / (p-1) > 2\sqrt{13}$
Numerically: $C(S4)$ in $[8.6290436642 \pm 1e-10]$
 $2\sqrt{13}$ in $[7.2111025509 \pm 1e-10]$

2. Causal Chain

This module depends on Module 4 (bound_10_4000.py stdout):

Parent SHA-256: b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed

The input file data/S14_primes.txt is byte-identical to Module 4 stdout:

```
$ ./bin/print_S14 | sha256sum
53315d4e6649a40b425edd445efbb937c0dec7a1aa571ea6b60f4f1033568387 - [chain intact]
```

Module 4 stdout format: 14 primes, comma-separated, single newline. arb_bost.py reads only the first 4 ($S4$) from this file.

3. Environment

Python 3.12 | mpmath 1.3.0 | 64 decimal places (~212 binary bits)
ARB 2.23.0: NOT AVAILABLE (NixOS sandbox)
Fallback: mpmath, exceeds 64-bit ARB precision
SHA-256 arb_bost.py : ed801430ee00e14eb799b31d68f3e2e94d35f468678a7126610d8f3dad3de57d
SHA-256 arb_bost.c : ad71a42d110791ed7594bd12780e755f5bf5d4e1d55ef48b9fdf5b5f5fc7fd3a

4. Actual Execution Output

```
$ python3 arb_bost.py data/S14_primes.txt 2>/dev/null
C(S4) in [1.4336768125 +/- 1.00e-10]
2*sqrt(13) in [7.2111025509 +/- 1.00e-10]
arb_gt(C, threshold) = 0
Exit code: 2
```

Stdout SHA-256:

88a6af4a36d4ff0885c4feb08f30382e5059f72520e202740966e2d5d5402c15 -

5. Forensic Analysis: Term-by-Term Proof

Using mpmath at 64 decimal places, the literal formula gives:

$p=2: \log(2)/(2-1) = 0.6931471805599453$
 $p=3: \log(3)/(3-1) = 0.5493061443340549$
 $p=19: \log(19)/(19-1) = 0.1635799432870245$
 $p=191: \log(191)/(191-1) = 0.0276435443581402$

C(S4) TOTAL = 1.4336768125391649

Comparison:

Computed C(S4) = 1.4336768125 (this work)
 Claimed C(S4) = 8.6290436642 (supervisor LaTeX)
 $2 \cdot \sqrt{13}$ = 7.2111025509
 Discrepancy = 7.1953668517

MATHEMATICAL PROOF of impossibility: the formula $\sum \log(p)/(p-1)$ over any 4 distinct primes is bounded above by the 4 smallest primes {2,3,5,7}: $0.693+0.549+0.402+0.324 = 1.968$. The claimed value 8.629 exceeds this maximum by 6.66. No choice of 4 distinct primes can produce 8.629 with this formula.

6. Hypothesis Testing: What Formula Gives 8.6290436642?

Systematic search over natural variants:

$\sum \log(p) \cdot p / (p-1)$ = 11.4221 (not 8.629)
 $\sum \log(p) / (p-1)^2$ = 0.9771 (not 8.629)
 $\sum (\log p)^2 / (p-1)$ = 1.7108 (not 8.629)
 $\sum \log(p)/p$ = 0.8952 (not 8.629)
 $-\sum \log(1-1/p)$ = 1.1579 (not 8.629)
 $\sum \log(p) / (p^{(\pi/10)} - 1)$ = 8.6975 (close, but off by 0.068)
 $\sum \log(p) / (p^{(1/\pi)} - 1)$ = 8.5433 (off by 0.086)
 $\sum \log(p) / (p^{0.31599} - 1)$ = 8.6290 [MATCHES, $\alpha \sim 0.31599$]

Finding: the value 8.6290436642 is consistent with the formula $\sum \log(p)/(p^\alpha - 1)$ evaluated at $\alpha \sim 0.31599$. This is close to, but not equal to, $\pi/10 = 0.31416$ (which gives 8.6975). The constant 0.31599 is not a recognizable mathematical constant.

Alternate hypothesis: the value 8.629 may be a computational error in the LaTeX draft, and the correct value is 1.434 (literal formula) or 8.697 (if the denominator should be $p^{(\pi/10)} - 1$, reflecting the fractional part of $\alpha_0 = 299 + \pi/10$).

7. Cryptographic Binding (SHA-256)

Artifact	SHA-256
arb_bost.py (fallback, S4)	ed801430ee00e14eb799b31d68f3e2e94d35f468678a7126610d8f3dad3de57d
arb_bost.c (source record)	ad71a42d110791ed7594bd12780e755f5bf5d4e1d55ef48b9fdf5b5f5fc7fd3a
data/S14_primes.txt (input)	53315d4e6649a40b425edd445efbb937c0dec7a1aa571ea6b60f4f1033568387
Stdout log (audit, exit 2)	88a6af4a36d4ff0885c4feb08f30382e5059f72520e202740966e2d5d5402c15
Module 4 parent (PARENT_SHA)	b810a7a331e47066e3eb4765a5ffdc17c1a56ddbff855a096c18ce2e9e2a19ed

8. Reproduction Commands

```
# Verify input (Module 4 stdout chain):
$ ./bin/print_S14 | sha256sum
53315d4e6649a40b425edd445efbb937c0dec7a1aa571ea6b60f4f1033568387 -

# Hash source:
$ sha256sum arb_bost.py
ed801430ee00e14eb799b31d68f3e2e94d35f468678a7126610d8f3dad3de57d arb_bost.py

# Run and hash stdout:
$ python3 arb_bost.py data/S14_primes.txt 2>/dev/null | sha256sum
88a6af4a36d4ff0885c4feb08f30382e5059f72520e202740966e2d5d5402c15 -
```

9. Required Supervisor Action

Module 5 bridges the discrete S4 certification (Module 4) and the GRH bound (Module 6). Certification is halted at this module.

Causal chain:

M1 (α_0) $\rightarrow$ M2 (κ) $\rightarrow$ M3 (CF $\pi/10$) $\rightarrow$ M4 (S14/S4)
 $\rightarrow$ M5 [HALTED] $\rightarrow$ M6 $\rightarrow$ M7 (manifest)

Please supply one of:

- (A) The Python or Sage snippet that computes $C(S4) = 8.6290436642$.
(The literal formula $\sum \log(p)/(p-1)$ gives 1.4337, not 8.629.)
- (B) Confirmation that the denominator should be $p^{(\pi/10)}-1$
instead of $p-1$. That formula gives 8.6975, still not 8.629.
- (C) The DOI or theorem number in the source paper where $C(S4)$
is defined and the value 8.629 is established.
- (D) Confirmation that the CORRECT $C(S4) = 1.4337$ and the
comparison threshold should NOT be $2\sqrt{13} = 7.211$.

This document is a Formula Audit Report under Battle Plan v1.6. All SHAs are real. No values are fabricated. Module 5 certification resumes upon supervisor-provided correction.